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Task graphs

Top-down and bottom-up approaches

Computing π

Parallel Algorithm Design

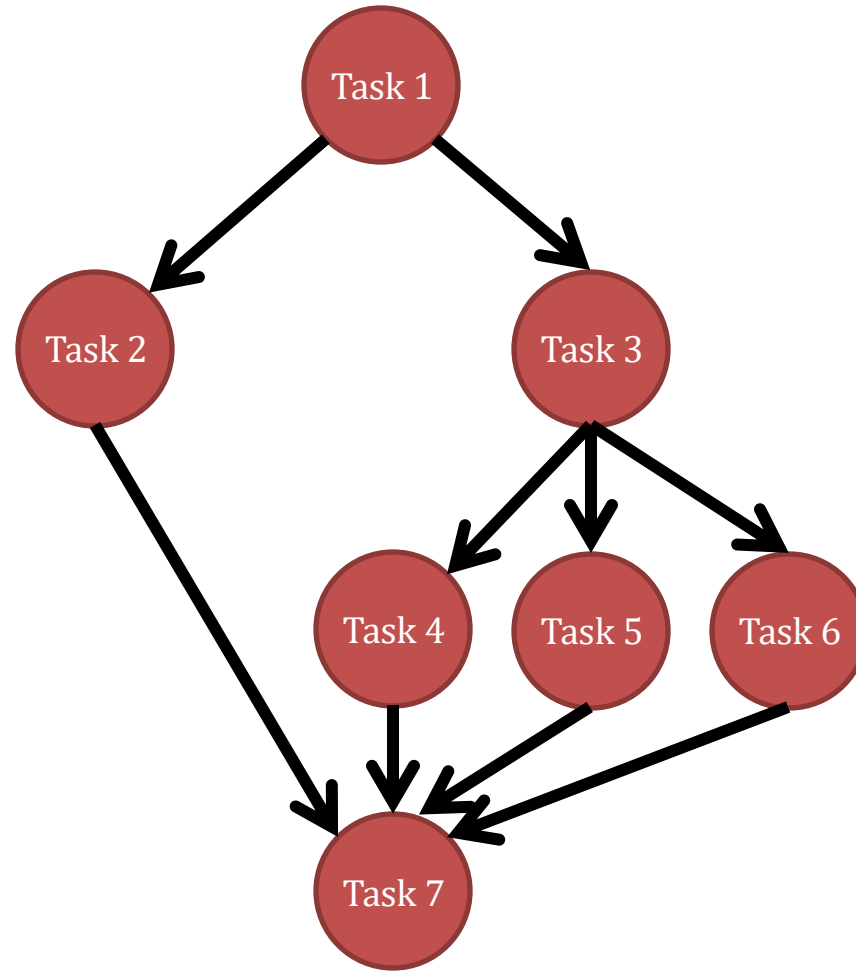
Abstract Model for Parallel Programs

Tasks: sequential program segments
with local data

Issues

- Concurrency
- Communication
- Synchronization
- Mapping
- Resources

Task Graph

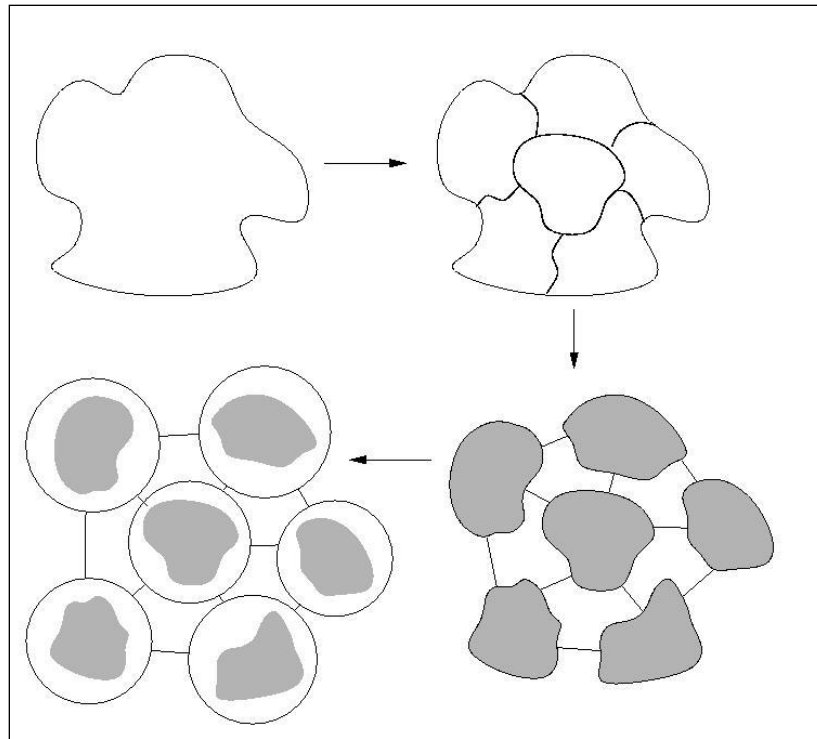


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Top Down Approach

- Partition problem into coarse-grained tasks
- Identify communication requirements
- Assign processors to tasks

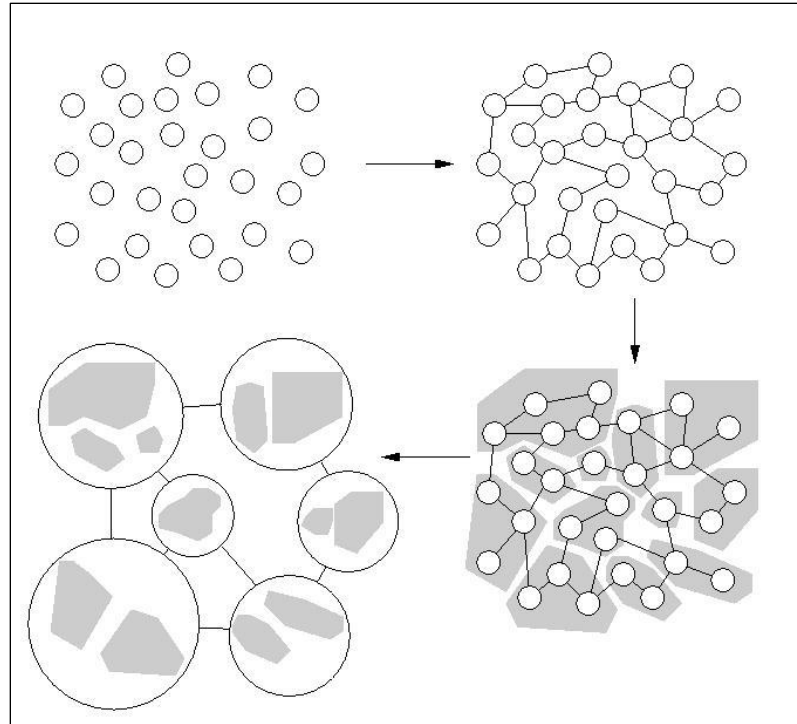
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Bottom Up Approach

- Identify fine-grained tasks
- Identify communication requirements
- Group tasks into coarse-grained tasks
- Assign processors to tasks

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Adding numbers in parallel

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
1	7	1	6	6	2	7	9	2	1	6	7	1	9	9	5	8	4	4	2	2	4	1	9	5	2	6	1	3	9	3	141
2	9	7	2	4	6	6	4	3	3	4	6	1	6	3	2	5	3	8	6	8	4	8	7	2	1	9	6	4	8	5	150
3	4	1	8	1	8	7	5	1	2	1	4	4	4	8	9	1	2	4	4	3	8	6	1	1	7	5	4	3	1	1	118
4	2	5	4	9	8	2	3	8	2	2	5	7	6	5	7	5	3	5	9	4	3	3	7	6	1	4	1	6	8	4	144
5	5	5	9	5	9	7	5	3	4	5	7	8	8	9	8	9	5	3	7	2	3	9	7	5	9	7	1	2	5	4	175
6	7	7	3	7	1	9	5	2	9	8	6	7	1	9	5	4	7	9	7	4	8	7	2	5	3	5	6	1	8	2	164
7	2	6	4	4	5	7	4	3	5	4	7	1	8	6	4	7	5	3	9	3	9	2	8	2	8	3	1	9	7	8	154
8	5	9	9	6	4	1	6	1	7	1	9	8	7	8	3	5	2	1	4	2	3	6	2	5	3	6	6	3	7	6	145
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11	4	3	4	8	9	3	2	5	8	8	8	1	8	5	9	1	3	9	1	2	1	9	2	4	3	1	5	8	7	3	144
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14	5	6	1	2	7	9	6	8	2	6	7	4	6	1	3	2	6	8	4	4	7	3	6	1	7	7	8	7	4	7	154
15	8	1	4	8	5	1	4	1	4	3	5	7	2	2	9	6	6	5	2	5	6	9	3	3	9	4	9	9	2	3	145
16	3	6	2	7	7	1	3	8	9	7	4	4	8	4	3	5	4	8	7	6	5	5	4	8	1	4	4	4	6	7	154
17	5	8	2	6	2	1	3	5	4	7	4	6	4	3	1	3	3	3	5	5	4	9	5	1	3	9	5	6	5	4	131
18	3	2	4	6	7	7	2	3	6	4	4	9	2	9	1	6	8	8	1	3	5	1	9	3	9	1	4	4	1	9	141
19	6	8	5	2	7	1	9	1	2	5	7	9	5	7	9	4	2	2	6	8	7	2	5	6	1	5	9	1	4	7	152
20	6	6	5	6	6	6	1	5	6	1	6	1	6	7	2	3	9	9	3	6	3	4	8	3	7	6	1	7	5	2	146
21	7	2	9	8	2	7	4	5	4	5	8	1	2	7	9	6	5	7	4	1	5	7	3	7	1	6	3	9	7	3	154
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23	5	9	5	1	8	1	9	3	5	9	8	5	3	1	9	1	1	5	5	2	5	7	3	8	8	5	4	7	6	5	153
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27	6	1	6	5	8	4	5	2	2	5	6	4	9	5	5	1	7	2	4	3	5	5	7	2	4	7	7	9	6	4	146
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30	5	6	1	6	4	3	9	5	5	7	7	2	9	5	6	8	5	6	5	6	2	2	8	4	9	4	2	4	6	8	159
	159	157	151	147	155	141	159	125	129	138	184	141	167	172	177	146	137	149	153	119	148	145	169	154	153	133	133	148	168	139	4496

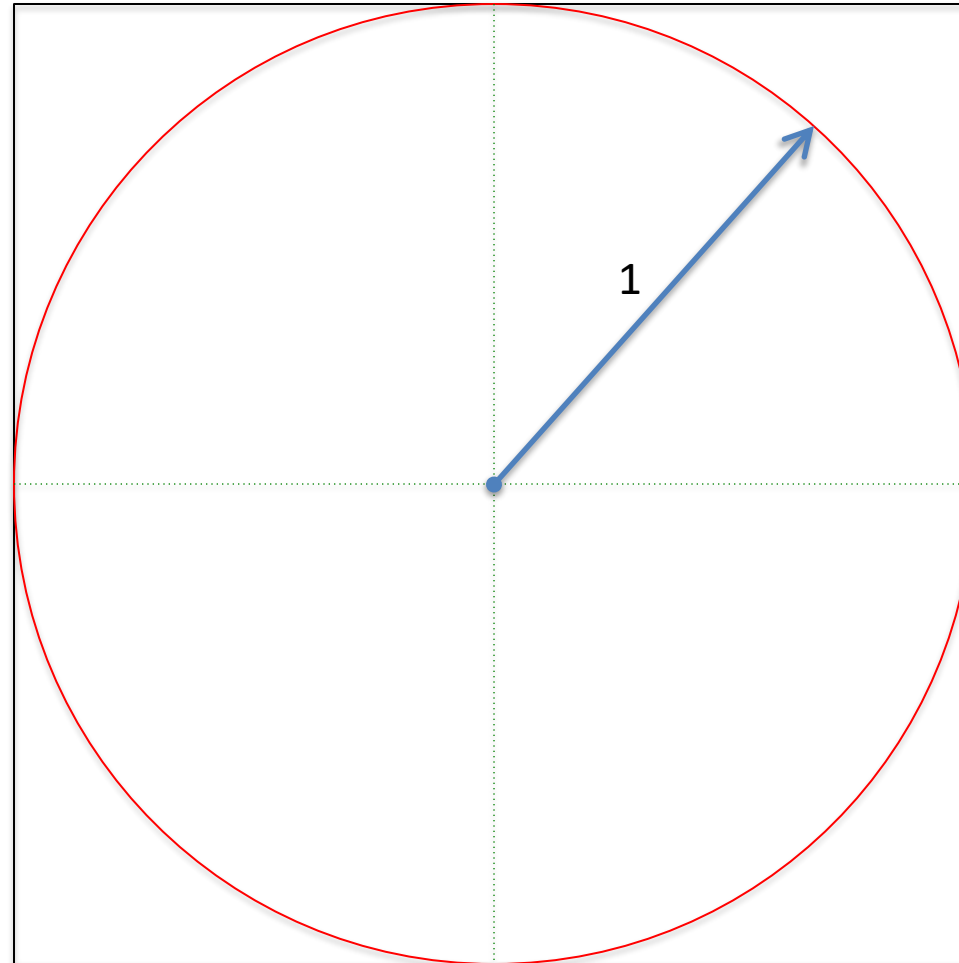
Issues Affecting Parallelism

- Partitioning
 - Load balance
 - Granularity
 - Static vs. Dynamic
 - Number of tasks
- Communication
 - Structured vs. unstructured
 - Local vs. global
 - Asynchronous vs. synchronous
 - Static vs. Dynamic
 - Concurrent
 - Overlap with computation



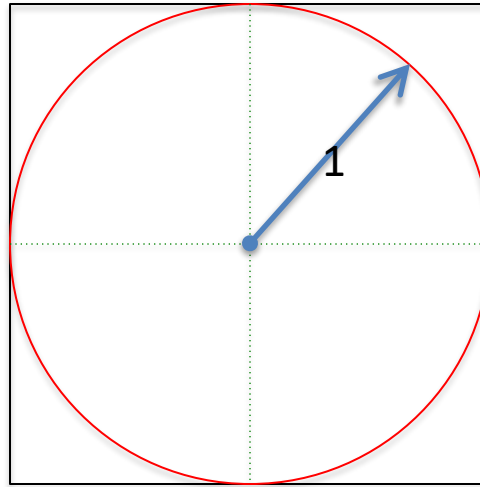
Compute π

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Algorithm to Compute π

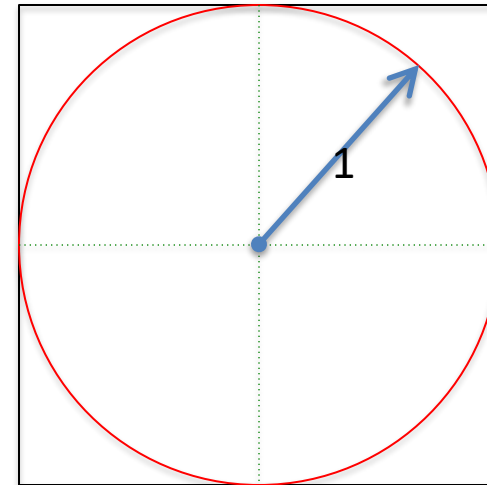
- Generate random points (x,y) in a unit length square
- Keep track of points that fall within a circle of unit diameter
- Since ratio of area of circle and area of square equals $\pi/4$ the fraction of points falling within the circle approaches $\pi/4$ for large number of random trials



Algorithm to Compute π ...

- Parallelization: bottom-up or top-down?
- Partitioning strategy
 - Number of tasks
 - Granularity of task
 - Load balance among processors
 - Mapping of tasks to processors: static or dynamic
- Communication among processors
 - Structured or unstructured
 - Local or global
 - Asynchronous or synchronous
 - Static or dynamic
 - Concurrent?
 - Overlap communication with computation?
- Accuracy of computation

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Another Algorithm to Compute π

- Compute the area under the curve:

$$f(x) = \frac{1}{1 + x^2}$$

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